

DEPARTMENT OF MATHEMATICS
UNIVERSITY OF MARYLAND
GRADUATE WRITTEN EXAM

August 2022

ALGEBRA (Ph.D. Version)

Instructions to the student

- a. Answer all six questions; each will be assigned a grade from 0 to 10.
 - b. Use a different booklet for each question. Write the problem number and your **code number (not your name)** on the outside of the booklet.
 - c. Keep scratch work on separate pages in the same booklet.
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1. (a) Let p be a prime and let G be a finite group whose order is p^n for some $n \geq 1$. Suppose G acts on a finite set X , and the cardinality of X is a multiple of p . Show that the number of fixed points of this action is a multiple of p .
(b) Suppose $H \neq 1$ is a normal subgroup of G . Let Z be the center of G . Show that $H \cap Z \neq 1$.
(c) Suppose $H \neq 1$ is a normal subgroup of G . Show that there exists a normal subgroup K of G such that $K \subset H$ and $[H : K] = p$.
2. Let $V \neq 0$ be a finite-dimensional vector space over a field F and let $T : V \rightarrow V$ be a linear transformation. A cyclic vector for T is a vector $v \in V$ such that $\{v, Tv, T^2v, \dots\}$ spans V . You may assume (without proving it) the fact that there exists a cyclic vector for T if and only if the minimal polynomial of T equals the characteristic polynomial of T .
(a) Suppose that every nonzero vector $v \in V$ is a cyclic vector for T . Show that the characteristic polynomial of T must be irreducible over F .
(b) Suppose that the characteristic polynomial of T is irreducible over F . Show that every nonzero $v \in V$ is a cyclic vector for T . (*Hint:* For a subspace W of V such that $T(W) \subseteq W$, the minimal polynomial of T restricted to W divides the minimal polynomial of T on V . (You do not need to prove this.))
3. Let $\mathbb{C}[x, y]$ be the ring of polynomials in 2 variables over the complex numbers and let (xy) be the ideal generated by xy . Find all prime ideals of $\mathbb{C}[x, y]/(xy)$. You should describe these prime ideals explicitly in terms of their generators.
4. Let R be a commutative ring with 1. Recall that an R -module I is injective if it satisfies the following condition: Whenever $f : A \rightarrow B$ is an injective homomorphism of R -modules and $p : A \rightarrow I$ is a homomorphism, then there exists a homomorphism $h : B \rightarrow I$ such that $hf = p$.
(a) (3 points) Show that the direct sum of two injective modules is injective.

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(b) (7 points) Suppose that we have a diagram of modules

$$\begin{array}{ccccccc}
 & & 0 & & 0 & & \\
 & & \downarrow & & \downarrow & & \\
 0 & \longrightarrow & A & \xrightarrow{f} & B & \xrightarrow{g} & C \longrightarrow 0 \\
 & & \downarrow \alpha & & & & \downarrow \gamma \\
 & & I_1 & & & & I_2
 \end{array}$$

where the row is exact, where I_1 and I_2 are injective, and where α and γ are injective. Show that there exists an injective module P , homomorphisms r and s , and an injection β such that the following diagram has exact rows and commutes:

$$\begin{array}{ccccccc}
 & & 0 & & 0 & & 0 \\
 & & \downarrow & & \downarrow & & \downarrow \\
 0 & \longrightarrow & A & \xrightarrow{f} & B & \xrightarrow{g} & C \longrightarrow 0 \\
 & & \downarrow \alpha & & \downarrow \beta & & \downarrow \gamma \\
 0 & \longrightarrow & I_1 & \xrightarrow{r} & I & \xrightarrow{s} & I_2 \longrightarrow 0
 \end{array}$$

You should prove that the bottom row is exact, that the diagram commutes, and that β is injective.

5. Let $f(x) = x^4 - 4x^2 + 1 \in \mathbb{Q}[x]$. Note that if α is a root of $f(x)$, then so are $\pm 1/\alpha$ and $-\alpha$.
 - (a) Show that $f(x)$ is irreducible.
 - (b) Let α be a root of $f(x)$. Show that $\mathbb{Q}(\alpha)/\mathbb{Q}$ is a Galois extension.
 - (c) Show that there are three distinct subfields K_1, K_2, K_3 of $\mathbb{Q}(\alpha)$ such that $[K_i : \mathbb{Q}] = 2$ for each i .
6. (a) Let G be a finite nonabelian simple group (that is, G has no nontrivial normal subgroups). Let $\rho : G \rightarrow \mathrm{GL}_2(\mathbb{C})$ be a two-dimensional representation of G . Show that ρ is either irreducible or the sum of two trivial representations.
 - (b) It is known that the only finite subgroups of $\mathrm{GL}_2(\mathbb{C})/\mathbb{C}^*$ (where $\mathbb{C}^*$ denotes the multiplicative group of nonzero scalar multiples of the identity) are isomorphic to subgroups of one of the following: D_n , A_5 , S_4 (where D_n is the dihedral group of order $2n$, S_n is the group of permutations of n objects, and A_n is the subgroup of even permutations; you may use the fact that D_n and S_4 are solvable groups). Use this fact to prove that if G is a nonabelian finite simple group with a nontrivial two-dimensional representation over $\mathbb{C}$, then $G = A_5$. (Note: You do not need to prove that A_5 has such a representation.)